

A note on solving a linear expectation model (preliminary)

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1 The problem

Assume there is a model of the form:

$$E_t \{f_\theta(y_{t+1}, y_t, y_{t-1}, \varepsilon_t)\} = 0, \quad (1)$$

where y_t is a vector of endogenous variables, θ are the structural parameters and ε_t a vector of exogenous shocks.

In a stochastic context, our model verifies three conditions:

$$E[\varepsilon_t] = 0, \quad (2)$$

$$E[\varepsilon_t \varepsilon_t'] = \Sigma_\varepsilon, \quad (3)$$

$$E[\varepsilon_t \varepsilon_\tau'] = 0 \text{ for } t \neq \tau, \quad (4)$$

which mean that (i) the mean of the shocks is zero because they are drawn from a Gaussian distribution; (ii) there is a covariance matrix Σ_ε which can either calibrated or estimated; (iii) shocks are not auto-correlated over time.

In a stochastic framework, the unknowns are the decision functions, i.e. how endogenous variables change over time:

$$y_t = g(y_{t-1}, \varepsilon_t) \quad (5)$$

The key problem is the way we approximate Equation 5. Usually, we make a linear approximation around a fixed point, however alternative solution methods exist to approximate Equation 5 (google global methods if you want to know more about these). A linear approximation of Equation 5, denoted $\hat{g}(\cdot)$, using a first order Taylor expansion is given by:

$$y_t = \hat{g}(y_{t-1}, \varepsilon_t) \quad (6)$$

$$= \bar{y} + A(y_{t-1} - \bar{y}) + B\varepsilon_t \quad (7)$$

where $\bar{y}$ is the fixed point (*i.e.* the steady state) of the approximation and A - B are matrices filled with first order derivatives. This approximation allows us to solve easily a rational expectation model. Dynare automatically computes these matrices, however it is possible to derive A - B by hand using a log-linearisation method.

2 A simple illustration

To illustrate how Dynare works in a linear environment, I present here a way to solve a linear expectation model by hand. Dynare employs the Schur decomposition to solve a model using perturbation methods. Here, I simply present a one equation model and approximate its policy function by hand. The solution provided is exactly the same as the one given by Dynare.

First, suppose that our economy is modelled by an hybrid new Keynesian Phillips curve given by:

$$\pi_t = \gamma\beta E_t \{\pi_{t+1}\} + (1 - \gamma)\pi_{t-1} + \eta_t \quad (8)$$

where π_t is the inflation rate, which is a linear function of inflation expectations $E_t \{\pi_{t+1}\}$ and of inflation indexation π_{t-1} . The weight for forward inflation is γ and $1-\gamma$ for backward inflation and β is the discount factor. Finally, η_t is a random Gaussian shock $N(0, \sigma^2)$ where σ is the standard deviation of the shock.

Recall that this model is already written in log-deviations, which means that the steady state is zero $\bar{\pi} = 0$ and the model is already linear. The policy function Equation 6 can thus be directly expressed as:

$$\pi_t = A\pi_{t-1} + B\eta_t. \quad (9)$$

We can thus combine Equation 9 in Equation 8 to get rid of expectations:

$$\begin{aligned} -\pi_t + \gamma\beta (A\pi_t + BE_t \{\eta_{t+1}\}) + (1 - \gamma)\pi_{t-1} + \eta_t &= 0 \\ -\pi_t + \gamma\beta A\pi_t + (1 - \gamma)\pi_{t-1} + \eta_t &= 0 \\ (\gamma\beta A - 1)\pi_t + (1 - \gamma)\pi_{t-1} + \eta_t &= 0 \end{aligned}$$

Agents use all information available now, which implies that they have no information about the realization of future shocks and then expect $E_t \{\eta_{t+1}\} = 0$, which explains why the term $\gamma B E_t \{\eta_{t+1}\}$ was dropped. Using again the trick to get rid of π_t :

$$\begin{aligned} (\gamma\beta A - 1) [A\pi_{t-1} + B\eta_t] + (1 - \gamma)\pi_{t-1} + \eta_t &= 0 \\ [(\gamma\beta A - 1) A + (1 - \gamma)] \pi_{t-1} + [1 + (\gamma\beta A - 1) B] \eta_t &= 0 \end{aligned}$$

For clarity purpose, we rewrite the previous expression as:

$$C\pi_{t-1} + D\eta_t = 0 \quad (10)$$

where $C = [(\gamma\beta A - 1) A + (1 - \gamma)]$ and $D = [1 + (\gamma\beta A - 1) B]$.

The solution which solves the previous expression is given by:

$$C = D = 0 \quad (11)$$

Which allows us to compute A and B :

$$0 = (\gamma\beta A^2 - A) + (1 - \gamma) \quad (12)$$

$$B = (1 - \gamma\beta A)^{-1} \quad (13)$$

The value of B is straightforward to compute, however the value of A is trickier as it is a quadratic expression. The closed form solution for A is:

$$A_1 = \frac{1 - \sqrt{1 - 4\theta\beta(1 - \theta)}}{2\theta\beta} \quad \text{or} \quad A_2 = \frac{1 + \sqrt{1 - 4\theta\beta(1 - \theta)}}{2\theta\beta} \quad (14)$$

Another key aspect, in addition to the type of approximation, is the solution to pick up. Here, we see that our economy allows to possible paths. In the profession, we always choose the stable one.

3 Numerical application

Suppose that the discount factor is $\beta = 0.99$ and that firms are forward looking with $\theta = 0.9$, it is possible to compute the policy function $g(\cdot)$. We find that: $A_1 = 0.1110$ and $A_2 = 1.0114$. The second solution is clearly instable as a small shock will make the inflation path explosive. So we naturally pick up A_1 as our solution and find that $B = 1.1097$. The fact that $B > 1$ is not worrying as it has no implication on the stability of the backward looking process described in [Equation 9](#). We also assume a 1% standard deviation for the shock with $\sigma = 0.01$.

To sum up, our policy function reads as:

$$\pi_t = 0.1110\pi_{t-1} + 1.1097\eta_t. \quad (15)$$

When you solve the model with Dynare, the latter automatically plots the policy function of the model which is exactly the same as our :

POLICY AND TRANSITION FUNCTIONS	
pi	
pi(-1)	0.110973
e	1.109726

3.1 Impulse response function

It is possible to compute the response of the model following the realization of one shock once, which we call Impulse Response Function (IRF, hereafter). This exercise allows economists to evaluate the propagation mechanism to the economy incurred by a single shock such as technology, spending, etc... And inflation for our case. We then usually compare the different propagation mechanisms under alternative calibrations, to see how the model is sensible to any set of parameters. For instance for monetary policy, we can evaluate the propagation of the model under different calibration for the interest rate reaction to inflation.

To illustrate here how we compute an IRF, I provide here a step by step example. At the beginning in the first period, our economy is at steady state with no fluctuation ($\pi_1 = 0$) and no shock ($\eta_1 = 0$). In a second period, we assume that our economy experiences an average 1% inflation shock through $\eta_2 = 0.01$, using [Equation 15](#), we find that

$$\begin{aligned} \pi_2 &= 0.1110\pi_1 + 1.1097\eta_2 \\ &= 0.1110 \times 0 + 1.1097 \times 0.01 \\ &= 0.0111 \end{aligned}$$

So basically for a 1% shock to inflation, we obtain a 1.11% inflation rise. We can thus proceed step by step to get the dynamic of inflation going back to steady state. The results are stored in the next table:

time t=	1	2	3	4	5
η_t	0	0.01	0	0	0
π_t	0	0.0111	0.0012	0.0001	0.0000

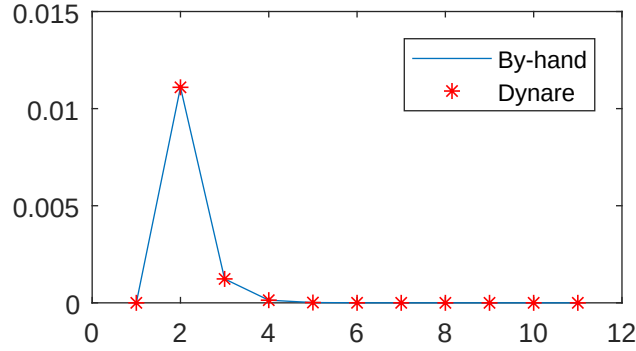


Figure 1: IRF of the model under the solution computed by hand and by Dynare

3.2 Artificial series

It also critical to generate artificial series, i.e. random business cycles. This allows economists to generate random fluctuations and get asymptotic moments of the model using a very large number of draws. This moments are second moments statistics such as variance and autocorrelation.

To simulate random series, we just need to pick up a random serie of draws taken from a normal distribution with a zero mean and a 0.01 standard deviation. For instance, a vector of random draws:

time t=	1	2	3	4	5	...
η_t	0	0.0089	-0.0115	-0.0107	-0.0081	...
π_t	0	0.0099	-0.0116	-0.0132	-0.0104	

Figure 2 plots some random business cycles generated by the random draws.

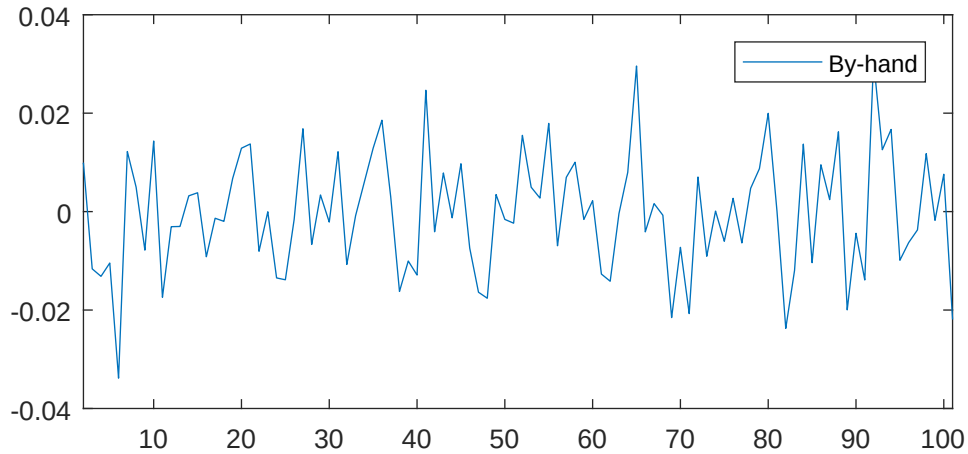


Figure 2: Artificial cycles of inflation

3.3 Second moment statistics

It is possible to compute second moment statistics through two methods:

1. **Asymptotic statistics:** basically the idea is to generate one (or more if you want uncertainty) large sample of artificial cycles, and compute from it the desired business cycle statistics.

2. **Stochastic calculus:** an alternative is to derive by hand a closed form expression of the variance. This clearly requires important knowledge in stochastic calculation. I illustrate here how we can obtain the variance of inflation from the policy function:

$$\begin{aligned} V[\pi_t] &= V[A\pi_{t-1} + B\eta_t] \\ &= V[A\pi_{t-1}] + V[B\eta_t] \\ &= A^2V[\pi_{t-1}] + B^2V[\eta_t] \end{aligned}$$

Recalling that $V[\pi_t] = V[\pi_{t-1}]$, then:

$$V[\pi_t] - A^2V[\pi_{t-1}] = B^2\sigma^2$$

Finally, the analytical expression of the variance of inflation is given by:

$$V[\pi_t] = \frac{B^2\sigma^2}{1 - A^2} \tag{16}$$

The variances/standard deviations through both artificial series and stochastic calculus provides the same value as the one provided by Dynare, i.e. $sd[\pi_t] = 0.0112$.